

ReflectionPad2d#

[https://pytorch.org/docs/stable/generated/torch.nn.ReflectionPad2d.html#torch](https://pytorch.org/docs/stable/generated/torch.nn.ReflectionPad2d.html#torch.nn.ReflectionPad2d)

Description:

Pads the input tensor using the reflection of the input boundary. For N-dimensional padding, use `torch.nn.functional.pad()`. padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 4-tuple, uses (padding_left\text{padding_left}padding_left, padding_right\text{padding_right}padding_right, padding_top\text{padding_top}padding_top, padding_bottom\text{padding_bottom}padding_bottom) Note that padding size should be less than the corresponding input dimension. Input: (N,C,Hin,Win)(N, C, H_{in}, W_{in})(N,C,Hin,Win) or (C,Hin,Win)(C, H_{in}, W_{in})(C,Hin,Win). Output: (N,C,Hout,Wout)(N, C, H_{out}, W_{out})(N,C,Hout,Wout) or (C,Hout,Wout)(C, H_{out}, W_{out})(C,Hout,Wout) where

Function Signature

```
class torch.nn.ReflectionPad2d(padding)[source]#
```

Parameters

Shape:

Code Examples

```
>>> m = nn.ReflectionPad2d(2)
>>> input = torch.arange(9, dtype=torch.float).reshape(1, 1, 3,
3)
>>> input
tensor([[[[0., 1., 2.],
          [3., 4., 5.],
          [6., 7., 8.]]]]])
>>> m(input)
tensor([[[[8., 7., 6., 7., 8., 7., 6.],
          [5., 4., 3., 4., 5., 4., 3.],
          [2., 1., 0., 1., 2., 1., 0.],
          [5., 4., 3., 4., 5., 4., 3.],
          [8., 7., 6., 7., 8., 7., 6.],
          [5., 4., 3., 4., 5., 4., 3.],
          [2., 1., 0., 1., 2., 1., 0.]]]]]])
>>> # using different paddings for different sides
>>> m = nn.ReflectionPad2d((1, 1, 2, 0))
>>> m(input)
tensor([[[[7., 6., 7., 8., 7.],
          [4., 3., 4., 5., 4.],
          [1., 0., 1., 2., 1.],
          [4., 3., 4., 5., 4.],
          [7., 6., 7., 8., 7.]]]]]])
```

Detailed Documentation

Documentation

Pads the input tensor using the reflection of the input boundary.

For N-dimensional padding, use `torch.nn.functional.pad()`.

padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 4-tuple, uses (padding_left\text{padding_left}padding_left, padding_right\text{padding_right}padding_right, padding_top\text{padding_top}padding_top, padding_bottom\text{padding_bottom}padding_bottom) Note that padding size should be less than the corresponding input dimension.

Input: (N,C,Hin,Win)(N, C, H_{in}, W_{in})(N,C,Hin,Win) or (C,Hin,Win)(C, H_{in}, W_{in})(C,Hin,Win).

Output: (N,C,Hout,Wout)(N, C, H_{out}, W_{out})(N,C,Hout,Wout) or (C,Hout,Wout)(C, H_{out}, W_{out})(C,Hout,Wout) where

$H_{out} = H_{in} + \text{padding_top} + \text{padding_bottom}$
 $H_{out} = H_{in} + \text{padding_top} + \text{padding_bottom}$

$W_{out} = W_{in} + \text{padding_left} + \text{padding_right}$
 $W_{out} = W_{in} + \text{padding_left} + \text{padding_right}$